t3-w05 — Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

Run `c0c3-prism-objective` · conditions C0, C1, C2, C3 · the pack shown is draw 3 of each.

HER LESSON PLAN

Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

Objectives

1. identify properties of cylinders and spheres
2. find the volume of a cube, a cuboid and a prism

Step 0 — intro: activating prior knowledge

Teacher recalls with the students what they have already been taught, so the work on cylinders, spheres and volume has something to stand on:

Step 1

Teacher displays shapes of cylinders and spheres while the students identify the properties. Teacher draws the shapes on the board and add to the properties already mentioned by the students if there is need.

- **teacher** displays shapes of cylinders and spheres
- **student** identify the properties
- **teacher** draws the shapes on the board and adds to the properties already mentioned by the students

1. 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

Step II

Teacher explains how to derive the formula for volume of cube and cuboid while the students listen and ask questions where they do not understand. They copy worked examples into their notes, which also work the volume of a prism as area of base times height.

- **teacher** explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height
- **student** listen and ask questions where they do not understand
- **student** copy worked examples into their notes

Evaluation

Teacher tests the students' understanding on the topic by giving them some questions to solve. Question: 1. Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. 2. Calculate the volume of cuboid with dimensions 5m by 2m by 3m. 3. What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm. 4. A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

1. Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
2. Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
3. What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
4. A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

HER LESSON NOTE

Verbatim, as transcribed. **C2 and C3 receive this; C0 and C1 never do** — which is the whole of what separates them.

Step 0 — intro: activating prior knowledge

```
1. properties of cubes and cuboid
2. properties of pyramids and cones
```

Step 1

```
#### Properties of a cylinder

**fig-01.** A cylinder drawn upright: an ellipse for the top face, two vertical sides
falling from its ends, and a curve closing the base.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
  \draw (-1.2,2.6) -- (-1.2,0);
  \draw (1.2,2.6) -- (1.2,0);
  \draw (-1.2,0) arc (180:360:1.2 and 0.35);
  \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
\end{tikzpicture}
\end{center}

1. It has two circular faces and one curved face
2. It has no vertex
3. It has two curved edges
4. It has three faces altogether

#### Properties of sphere

**fig-02.** A sphere drawn as a circle with an ellipse across its middle to show the
curved surface.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) circle (1.3);
  \draw (-1.3,0) arc (180:360:1.3 and 0.42);
  \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
\end{tikzpicture}
\end{center}

1. A sphere is symmetrical from all direction
2. It has no edge and no vertex
3. It has one curved surface

#### Student activity

1. How many circular faces does a solid cylinder have?
2. A cylinder has how many faces.
```

Step II

```
#### Volume of cube and cuboid

1. Volume of cuboid = Area of base $\times$ height

   $= \text{length} \times \text{breadth} \times \text{height}$

   $= lbh$

2. Volume of cube = Length $\times$ Length $\times$ Length

   $= l^{3}$

**Example 1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.

Solution:

Volume of cuboid $= L \times b \times h$

length $= 3cm$

breadth $= 2cm$

height $= 8cm$

Volume $= 3cm \times 2cm \times 8cm$

$= 48cm^{3}$

**Example 2.** The volume of a cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the
height of the cuboid.

Solution:

Volume of cuboid $= L \times b \times h$

length $= 6cm$

breadth $= 2cm$

height $= ?$

Volume $= 24cm^{3}$

$24cm^{3} = 6cm \times 2cm \times h$

$24cm^{3} = 12cm^{2}h$

Divide both side by 12cm$^{2}$

$\frac{24cm^{3}}{12cm^{2}} = \frac{12cm^{2}h}{12cm^{2}}$

$2cm = h$

$h = 2cm$

#### Volume of cube

**Example 3.** Calculate the volume of a cube of side 3cm.
```

```
Solution

Volume of cube $= l^{3}$

$V = (3cm)^{3}$

$V = 3cm \times 3cm \times 3cm$

$V = 27cm^{3}$

#### Volume of prism

Calculate the volume of the prism below.

**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a base
of 5 m and a height of 3 m, and the prism runs back 10 m.

\begin{center}
\begin{tikzpicture}[scale=0.42,line width=0.5pt]
  \draw (0,0) -- (5,0) -- (2.5,3) -- cycle;
  \draw (5,0) -- (8.5,1.8);
  \draw (2.5,3) -- (6,4.8);
  \draw (8.5,1.8) -- (6,4.8);
  \draw[dashed] (0,0) -- (3.5,1.8);
  \draw[dashed] (3.5,1.8) -- (8.5,1.8);
  \draw[dashed] (3.5,1.8) -- (6,4.8);
  \draw[dashed] (2.5,0) -- (2.5,3);
  \node[below] at (2.5,-0.15) {$5\,\mathrm{m}$};
  \node[left] at (2.35,1.5) {$3\,\mathrm{m}$};
  \node[below right] at (6.9,0.8) {$10\,\mathrm{m}$};
\end{tikzpicture}
\end{center}

Solution:

Area of triangular face $= \frac{1}{2}bh$

$= \frac{1}{2} \times 5 \times 3$

$= 7.5m^{2}$

Volume of prism $=$ Area of base $\times$ height

$= 7.5m^{2} \times 10m$

$= 75m^{3}$
```

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube, a cuboid and a prism

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — find the volume of a cube, a cuboid and a prism

`objective_text`

find the volume of a cube, a cuboid and a prism

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Cylinders, Spheres, and Finding Volume

EXPLANATION

Two solid shapes you meet all the time are the cylinder and the sphere. Let's look closely at what makes each one special, so you can pick them out anywhere.

A cylinder has two identical flat circular faces, one at the top and one at the bottom. These two circles are called the bases. The two bases are joined by a single curved surface that wraps all the way around. A cylinder has no sharp points (no vertices), but it does have two edges — the two circular rims where the curved surface meets each base. Because of that curved surface, a cylinder can roll along the ground, but only in a straight line (it rolls end over end, not in all directions). Think of a tin of baked beans or a drinking glass.

A sphere is perfectly round — every point on its surface is exactly the same distance from its centre. That fixed distance is called the radius. A sphere has no flat faces, no edges, and no vertices at all. Because it has only one continuous curved surface, a sphere rolls in every direction. Think of a football or a tennis ball.

Here is a quick way to tell them apart: a sphere is the only shape that rolls freely in every direction and has no flat face, no edge, and no corner. A cylinder has exactly two flat circular faces, two circular edges, and no corners.

EXPLANATION

Volume is the amount of space a solid takes up, and we measure it in cubic units such as cm^3 or m^3. Three box-like solids have a simple volume formula you can rely on.

A cube is a solid where every edge is the same length. If its edge is s, then its volume is:
Volume of cube = $s \times s \times s = s^3$

A cuboid (a rectangular box) has three different edges — length l, width w and height h. Its volume is:
Volume of cuboid = $l \times w \times h$

A prism is any solid that has the same shape all the way through from one end to the other. The two ends are identical polygons, and they are joined by flat rectangular faces. The volume of any prism is:
Volume of prism = area of the cross-section × length

The cross-section is the shape of the two identical ends — for example, if a prism has a triangular end, you first find the area of that triangle and then multiply by the prism's length. So a cuboid is actually a prism whose cross-section is a rectangle, and the rule "area of cross-section × length" gives the same result as $l \times w \times h$.

WORKED_EXAMPLE

Worked Example 1 — Properties of a cylinder
A tin of soup has two flat circular ends and one curved surface that joins them.
How many faces, edges and vertices does it have?
The two circular ends are flat faces, and the curved surface is not flat but it is still one surface. So there are 3 faces in total. There are two circular edges — the rim where the curved surface meets each circular base. There are no vertices, because no flat faces meet at a point.

Worked Example 2 — Volume of a cube
Find the volume of a cube with edge 4 cm.
Volume = $s \times s \times s = 4 \times 4 \times 4 = 64$ cm^3.

Worked Example 3 — Volume of a cuboid
A box is 5 cm long, 3 cm wide and 2 cm high. Find its volume.
Volume = $l \times w \times h = 5 \times 3 \times 2 = 30$ cm^3.

Worked Example 4 — Volume of a prism
A triangular prism has a triangular end whose area is 8 cm^2 and a length of 6 cm. Find its volume.
Volume = area of cross-section × length = $8 \times 6 = 48$ cm^3.

1. A ball has only one continuous curved surface and no flat faces, no edges and no vertices. Is the ball a cylinder or a sphere? Give a reason.

answer: It is a sphere, because a sphere has no flat faces, no edges and no vertices, and it rolls freely in every direction.

2. A drinking glass is shaped like a cylinder. State the number of flat faces, the number of edges and the number of vertices it has.
answer: 2 flat faces, 2 edges, 0 vertices.

3. Find the volume of a cube whose edge is 5 cm.
answer: 125 cm^3.

4. A cuboid is 8 cm long, 4 cm wide and 3 cm high. Find its volume.
answer: 96 cm^3.

5. A prism has a triangular cross-section with area 12 cm^2 and a length of 7 cm. Find its volume.
answer: 84 cm^3.

Objective 2

Volume of Cubes, Cuboids and Prisms

> EXPLANATION
> Volume is the amount of space a solid takes up. It is measured in cubic units, like cubic centimetres (cm^3) or cubic metres (m^3).
>
> A **cube** has every edge the same length. If the edge is a, then:
> $$\text{Volume} = a \times a \times a = a^3$$
>
> A **cuboid** (a rectangular box) has three different edge lengths: length l, width w and height h. Its volume is:
> $$\text{Volume} = l \times w \times h$$
>
> A **prism** is a solid with the same shape (a face) all the way through. A cube and a cuboid are both prisms. For any prism, the volume is found by multiplying the area of the end face (the cross-section) by the length (or height):
> $$\text{Volume} = \text{area of cross-section} \times \text{length}$$
>
> So the same rule works for a cube, a cuboid and any other prism: find the area of the end face, then multiply by how long the prism is.

> WORKED_EXAMPLE
> **Example 1: a cube**
> A cube has edges of length 5 cm. Find its volume.
>
> $$\text{Volume} = a^3 = 5 \times 5 \times 5 = 125$$
> The volume is **125 cm^3**.
>
> **Example 2: a cuboid**
> A cuboid is 8 cm long, 3 cm wide and 4 cm high. Find its volume.
>
> $$\text{Volume} = l \times w \times h = 8 \times 3 \times 4 = 96$$
> The volume is **96 cm^3**.
>
> **Example 3: a prism**
> A triangular prism has a triangular end face with area 12 cm^2 and is 7 cm long. Find its volume.
>
> $$\text{Volume} = \text{area of cross-section} \times \text{length} = 12 \times 7 = 84$$
> The volume is **84 cm^3**.

1. A cube has edges of length 6 cm. What is its volume?
answer: 216 cm^3

2. A cuboid is 10 cm long, 4 cm wide and 3 cm high. What is its volume?
answer: 120 cm^3

3. A prism has an end face with area 15 cm^2 and is 8 cm long. What is its volume?
answer: 120 cm^3

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube, a cuboid and a prism

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

`current_step`

1

`instructional_events`

• present_content
• elicit_performance
• present_content

`student_tasks`

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — find the volume of a cube, a cuboid and a prism

`objective_text`

find the volume of a cube, a cuboid and a prism

`current_step`

2

`instructional_events`

• present_content
• provide_learning_guidance

`assessment_points`

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Cylinders and Spheres: Getting to Know Their Shapes

> EXPLANATION
> Two shapes you see every day are the cylinder and the sphere. A tin of baked beans or a toilet roll is a cylinder; a football or an orange is a sphere. They are three-dimensional shapes, so they take up space, and each has its own special set of properties. Let us look at each one carefully.
>
> The cylinder
>
> Picture a tin of baked beans. A cylinder has two flat circular faces — one at the top and one at the bottom. These two circles are equal in size and lie parallel to each other. Between them runs a curved surface that joins the two circles, like the curved side of the tin. So a cylinder has:
>
> two circular faces (the top and the bottom), equal and parallel;
> one curved surface (the side);
> no vertices (no sharp corner points);
> two circular edges (the rims where each circular face meets the curved side).
>
>
> If you count all the surfaces, a cylinder has 2 flat circular faces and 1 curved surface — 3 surfaces in all.
>
> The sphere
>
> Now picture a football. A sphere is perfectly round — every point on its surface is the same distance from its centre. A sphere has no flat faces, no edges and no corners. It is just one single curved surface that curves all the way around, with no starting or ending point.

> WORKED_EXAMPLE
> Worked example
>
> Is this statement true or false? “A cylinder has two circular faces and one curved surface.”
>
> Answer: True. The two circular faces are the top and bottom of the cylinder, and the curved surface is the side that joins them. The statement describes the cylinder correctly.
>
> Worked example
>
> A ball has no flat faces, no edges and no vertices. Name the three-dimensional shape it is, and say how many surfaces it has.
>
> Answer: The ball is a sphere. It has just one surface — a single curved surface that wraps all the way around it.

1. A solid cylinder has two circular faces (the top and the bottom) and one curved surface. How many faces does a cylinder have altogether?

answer: A cylinder has 2 faces (the two circular faces — the top and the bottom). The curved side is a curved surface, not counted as a flat face.

2. How many circular faces does a solid cylinder have?

answer: A solid cylinder has 2 circular faces — one at the top and one at the bottom.

3. A sphere has no flat faces, no edges and no vertices. What is the name of its only surface?

answer: It is a curved surface — one single curved surface that wraps all the way around the sphere.

Objective 2

Finding the Volume of Cubes, Cuboids and Prisms

EXPLANATION

Volume tells you how much space a solid shape takes up. You measure it in cubic units, such as cubic centimetres (cm^3) or cubic metres (m^3). To find the volume of a cube or a cuboid, you multiply its length, its width and its height together.

Think of a cuboid as a stack of cubes. A cuboid that is 4 cm long, 2 cm wide and 3 cm high can be pictured as 3 layers, each layer holding 4 × 2 = 8 cubes. That gives 3 × 8 = 24 cubes, so its volume is 24 cm^3. What you have done is simply multiply the three dimensions:

Volume of a cuboid = length × width × height

A cube is just a cuboid whose length, width and height are all the same size. So if a cube has edges of length s, its volume is:

Volume of a cube = s × s × s = s^3

A prism is a solid with two identical, parallel faces (its bases) joined by flat rectangular sides. The volume of any prism is found by multiplying the area of one base by the height of the prism:

Volume of a prism = area of base × height

A cuboid is itself a prism whose base is a rectangle, so its base area is length × width, and multiplying by the height gives exactly the same formula as before.

WORKED_EXAMPLE

Worked example 1: the volume of a cuboid

Find the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.

Use Volume = length × width × height.

Volume = 8 × 4 × 3 = 32 × 3 = 96 cm^3.

So the cuboid takes up 96 cm^3 of space.

Worked example 2: the volume of a cube

Find the volume of a cube whose edges are each 4 cm long.

Use Volume = s^3 = s × s × s.

Volume = 4 × 4 × 4 = 16 × 4 = 64 cm^3.

So the cube takes up 64 cm^3 of space.

Worked example 3: the volume of a prism

A triangular prism has a base of area 12 cm^2 and a height of 5 cm. Find its volume.

Use Volume = area of base × height.

Volume = 12 × 5 = 60 cm^3.

So the prism takes up 60 cm^3 of space.

1. Find the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
answer: 96 cm^3

2. Find the volume of a cuboid with dimensions 5 m by 2 m by 3 m.

answer: 30 m^3

3. What is the volume of a cube of edges 4 cm × 4 cm × 4 cm?

answer: 64 cm^3

4. A rectangular tank is 6 m long and 2 m wide and holds 36 m^3 of water when full. How deep is the water in the tank?

answer: 3 m

C2 `C1 plus the lesson note as source evidence.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube, a cuboid and a prism

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

objective_text

identify properties of cylinders and spheres

current_step

1

instructional_events

• present_content
• elicit_performance
• present_content

student_tasks

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

source_figures


```
[
 {
  "caption": "A cylinder drawn upright: an ellipse for the top face, two vertical sides falling
from its ends, and a curve closing the base.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "A sphere drawn as a circle with an ellipse across its middle to show the curved
surface.",
  "figure_ref": "fig-02"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Properties of a cylinder\n\n**fig-01.** A cylinder drawn upright: an ellipse
for the top face, two vertical sides\nfalling from its ends, and a curve closing the base.
\n\n[drawing attached to this section as fig-01]\n\n1. It has two circular faces and one curved
face\n2. It has no vertex\n3. It has two curved edges\n4. It has three faces altogether\n\n####
Properties of sphere\n\n**fig-02.** A sphere drawn as a circle with an ellipse across its middle
to show the\ncurved surface.\n\n[drawing attached to this section as fig-02]\n\n1. A sphere is
symmetrical from all direction\n2. It has no edge and no vertex\n3. It has one curved
surface\n\n#### Student activity\n\n1. How many circular faces does a solid cylinder have?\n2. A
cylinder has how many faces."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 2 — find the volume of a cube, a cuboid and a prism

objective_text

find the volume of a cube, a cuboid and a prism

current_step

2

instructional_events

• present_content
• provide_learning_guidance

assessment_points

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

source_figures


```
[
 {
  "caption": "A triangular prism lying on its rectangular face. The triangular end has a base of
5 m and a height of 3 m, and the prism runs back 10 m.",
  "figure_ref": "fig-03"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Volume of cube and cuboid\n\n1. Volume of cuboid = Area of base $\\times$
height\n\n   $= \\text{length} \\times \\text{breadth} \\times \\text{height}$\n\n   $= lbh$
\n\n2. Volume of cube = Length $\\times$ Length $\\times$ Length\n\n   $= l^{3}$\n\n**Example
1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 3cm$\n\nbreadth $= 2cm$\n\nheight $=
8cm$\n\nVolume $= 3cm \\times 2cm \\times 8cm$\n\n$= 48cm^{3}$\n\n**Example 2.** The volume of a
cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the\nheight of the cuboid.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 6cm$\n\nbreadth $= 2cm$\n\nheight
$= ?$\n\nVolume $= 24cm^{3}$\n\n$24cm^{3} = 6cm \\times 2cm \\times h$\n\n$24cm^{3} = 12cm^{2}h$
\n\nDivide both side by 12cm$^{2}$\n\n$\\frac{24cm^{3}}{12cm^{2}} = \\frac{12cm^{2}h}{12cm^{2}}$
\n\n$2cm = h$\n\n$h = 2cm$\n\n#### Volume of cube\n\n**Example 3.** Calculate the volume of a
cube of side 3cm.\n\nSolution\n\nVolume of cube $= l^{3}$\n\n$V = (3cm)^{3}$\n\n$V = 3cm \\times
3cm \\times 3cm$\n\n$V = 27cm^{3}$\n\n#### Volume of prism\n\nCalculate the volume of the prism
below.\n\n**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a
base\nof 5 m and a height of 3 m, and the prism runs back 10 m.\n\n[drawing attached to this
section as fig-03, marked 5 m, 3 m, 10 m]\n\nSolution:\n\nArea of triangular face $= \\frac{1}{2}
bh$\n\n$= \\frac{1}{2} \\times 5 \\times 3$\n\n$= 7.5m^{2}$\n\nVolume of prism $=$ Area of base $
\\times$ height\n\n$= 7.5m^{2} \\times 10m$\n\n$= 75m^{3}$"
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Cylinders and Spheres: Shapes That Are Curved

> EXPLANATION
> Two three-dimensional shapes you meet all the time are the cylinder and the sphere. A drinks tin is a cylinder; a ball is a sphere. Even though they look quite different from the cubes and cuboids you already know, you can describe each one by its faces, edges and vertices.
>
> Look at the cylinder in fig-01. A cylinder stands on a circular base and has a circular top. That means it has two circular faces — one on the top and one on the bottom. Around the outside, between those two circles, runs a single smooth surface that curves all the way round: the curved face. So a solid cylinder has three faces altogether — two circular and one curved.
>
> Now count edges and vertices. The curved face meets the top circle along one edge and the bottom circle along another, so a cylinder has two curved edges. But where do the faces come together to make a point? They never do — the curved face simply wraps around. So a cylinder has no vertex.

> VISUAL
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
> \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
> \draw (-1.2,2.6) -- (-1.2,0);
> \draw (1.2,2.6) -- (1.2,0);
> \draw (-1.2,0) arc (180:360:1.2 and 0.35);
> \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
> \end{tikzpicture}

> EXPLANATION
> Now turn to the sphere in fig-02. A sphere is the shape of a ball — every point on its surface is the same distance from its centre. Because of that, a sphere looks identical from every direction: it is symmetrical from all directions.
>
> Unlike the cylinder, a sphere has no flat face at all. Its whole surface is one single curved surface. And because there is nothing flat and no line where faces meet, a sphere has no edge and no vertex.

> VISUAL
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
> \draw (0,0) circle (1.3);
> \draw (-1.3,0) arc (180:360:1.3 and 0.42);
> \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
> \end{tikzpicture}

1. 1. How many circular faces does a solid cylinder have?
   answer: A solid cylinder has 2 circular faces — one on the top and one on the bottom.
2. 2. How many faces does a cylinder have altogether?
   answer: A cylinder has 3 faces altogether: 2 circular faces and 1 curved face.
3. 3. Does a sphere have any edges or vertices?
   answer: No. A sphere has no edge and no vertex — its whole surface is one curved surface.

Volume of a Cube, a Cuboid and a Prism

EXPLANATION

Volume is the amount of space a three-dimensional shape fills, and we measure it in cubic units such as cm^3 or m^3. For a cuboid, think of the base as being made of layers that are stacked up to the height, so the volume is found by multiplying its three dimensions together:

Volume of cuboid = length × breadth × height = $l \times b \times h$

A cube is a special cuboid whose three sides are all the same length, so we simply multiply the side by itself three times:

Volume of cube = length × length × length = l^3

WORKED_EXAMPLE

Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.

Volume of cuboid = $l \times b \times h$
$= 3cm \times 2cm \times 8cm$
$= 48cm^3$

WORKED_EXAMPLE

Sometimes you know the volume and the base, and you need to find the height. The volume of a cuboid is $24cm^3$ and its base is 6cm by 2cm. Find its height.

Volume = $l \times b \times h$
$24cm^3 = 6cm \times 2cm \times h$
$24cm^3 = 12cm^2 \times h$

Divide both sides by $12cm^2$:
$24cm^3 \div 12cm^2 = h$
$2cm = h$

So the height of the cuboid is 2cm.

WORKED_EXAMPLE

Calculate the volume of a cube of side 3cm.

Volume of cube = l^3
$= 3cm \times 3cm \times 3cm$
$= 27cm^3$

WORKED_EXAMPLE

A prism has a uniform cross-section, so its volume is found by multiplying the area of its base by the height (the distance it runs back). Look at the triangular prism in fig-03. Its end is a triangle with base 5m and height 3m, and the prism runs back 10m.

First find the area of the triangular face:
Area of triangle = $\frac{1}{2} \times b \times h = \frac{1}{2} \times 5m \times 3m = 7.5m^2$

Then find the volume:
Volume of prism = area of base × height
$= 7.5m^2 \times 10m$
$= 75m^3$

VISUAL

```
\begin{tikzpicture}[scale=0.42,line width=0.5pt]
  \draw (0,0) -- (5,0) -- (2.5,3) -- cycle;
  \draw (5,0) -- (8.5,1.8);
  \draw (2.5,3) -- (6,4.8);
  \draw (8.5,1.8) -- (6,4.8);
  \draw[dashed] (0,0) -- (3.5,1.8);
  \draw[dashed] (3.5,1.8) -- (8.5,1.8);
  \draw[dashed] (3.5,1.8) -- (6,4.8);
```

```
  \draw[dashed] (2.5,0) -- (2.5,3);
  \node[below] at (2.5,-0.15) {$5\,\mathrm{m}$};
  \node[left] at (2.35,1.5) {$3\,\mathrm{m}$};
  \node[below right] at (6.9,0.8) {$10\,\mathrm{m}$};
\end{tikzpicture}
```

1. Calculate the volume of a cuboid that is 8cm long, 4cm wide and 3cm high.
answer: 96cm^3

2. Calculate the volume of a cuboid with dimensions 5m by 2m by 3m.
answer: 30m^3

3. What is the volume of a cube of side 4cm?
answer: 64cm^3

4. A rectangular tank that is 6m long and 2m wide holds 36m^3 of water. How deep is the water in the tank?
answer: 3m deep

C3 `C2 plus a validation and repair pass over the draft.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube, a cuboid and a prism

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

objective_text

identify properties of cylinders and spheres

current_step

1

instructional_events

• present_content
• elicit_performance
• present_content

student_tasks

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

source_figures


```
[
 {
  "caption": "A cylinder drawn upright: an ellipse for the top face, two vertical sides falling
from its ends, and a curve closing the base.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "A sphere drawn as a circle with an ellipse across its middle to show the curved
surface.",
  "figure_ref": "fig-02"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Properties of a cylinder\n\n**fig-01.** A cylinder drawn upright: an ellipse
for the top face, two vertical sides\nfalling from its ends, and a curve closing the base.
\n\n[drawing attached to this section as fig-01]\n\n1. It has two circular faces and one curved
face\n2. It has no vertex\n3. It has two curved edges\n4. It has three faces altogether\n\n####
Properties of sphere\n\n**fig-02.** A sphere drawn as a circle with an ellipse across its middle
to show the\ncurved surface.\n\n[drawing attached to this section as fig-02]\n\n1. A sphere is
symmetrical from all direction\n2. It has no edge and no vertex\n3. It has one curved
surface\n\n#### Student activity\n\n1. How many circular faces does a solid cylinder have?\n2. A
cylinder has how many faces."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 2 — find the volume of a cube, a cuboid and a prism

objective_text

find the volume of a cube, a cuboid and a prism

current_step

2

instructional_events

• present_content
• provide_learning_guidance

assessment_points

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

source_figures


```
[
 {
  "caption": "A triangular prism lying on its rectangular face. The triangular end has a base of
5 m and a height of 3 m, and the prism runs back 10 m.",
  "figure_ref": "fig-03"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Volume of cube and cuboid\n\n1. Volume of cuboid = Area of base $\\times$
height\n\n   $= \\text{length} \\times \\text{breadth} \\times \\text{height}$\n\n   $= lbh$
\n\n2. Volume of cube = Length $\\times$ Length $\\times$ Length\n\n   $= l^{3}$\n\n**Example
1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 3cm$\n\nbreadth $= 2cm$\n\nheight $=
8cm$\n\nVolume $= 3cm \\times 2cm \\times 8cm$\n\n$= 48cm^{3}$\n\n**Example 2.** The volume of a
cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the\nheight of the cuboid.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 6cm$\n\nbreadth $= 2cm$\n\nheight
$= ?$\n\nVolume $= 24cm^{3}$\n\n$24cm^{3} = 6cm \\times 2cm \\times h$\n\n$24cm^{3} = 12cm^{2}h$
\n\nDivide both side by 12cm$^{2}$\n\n$\\frac{24cm^{3}}{12cm^{2}} = \\frac{12cm^{2}h}{12cm^{2}}$
\n\n$2cm = h$\n\n$h = 2cm$\n\n#### Volume of cube\n\n**Example 3.** Calculate the volume of a
cube of side 3cm.\n\nSolution\n\nVolume of cube $= l^{3}$\n\n$V = (3cm)^{3}$\n\n$V = 3cm \\times
3cm \\times 3cm$\n\n$V = 27cm^{3}$\n\n#### Volume of prism\n\nCalculate the volume of the prism
below.\n\n**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a
base\nof 5 m and a height of 3 m, and the prism runs back 10 m.\n\n[drawing attached to this
section as fig-03, marked 5 m, 3 m, 10 m]\n\nSolution:\n\nArea of triangular face $= \\frac{1}{2}
bh$\n\n$= \\frac{1}{2} \\times 5 \\times 3$\n\n$= 7.5m^{2}$\n\nVolume of prism $=$ Area of base $
\\times$ height\n\n$= 7.5m^{2} \\times 10m$\n\n$= 75m^{3}$"
 }
]
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Cylinders and Spheres: Their Properties

EXPLANATION

In this section you will learn to identify the properties of two solid shapes: the cylinder and the sphere. You already know the cube, cuboid, pyramid and cone; now we look at shapes whose faces curve.

A **cylinder** looks like a tin can or a drum. Look at the cylinder in fig-01: it has a circular top, a circular base, and a curved side joining them. From the figure you can read off its properties:

1. It has **two circular faces** (the top and the base) and **one curved face**.
2. It has **no vertex** (no corner where edges meet).
3. It has **two curved edges** (the circles where the curved face meets the top face and the base).
4. Altogether it has **three faces**.

A **sphere** looks like a ball. Look at the sphere in fig-02. Its properties are:

1. It is **symmetrical from all directions** — turn it any way and it looks the same.
2. It has **no edge** and **no vertex**.
3. It has **one curved surface**.

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) circle (1.3);
  \draw (-1.3,0) arc (180:360:1.3 and 0.42);
  \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
  \draw (-1.2,2.6) -- (-1.2,0);
  \draw (1.2,2.6) -- (1.2,0);
  \draw (-1.2,0) arc (180:360:1.2 and 0.35);
  \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
\end{tikzpicture}
```

WORKED_EXAMPLE

Let us compare a cylinder and a sphere side by side.

Example. Count the faces, edges and vertices of a cylinder.

In fig-01 you can see the two circular ends and the curved side between them. So the cylinder has:

- Faces: two circular faces plus one curved face = **3 faces**
- Edges: two curved edges (where the curved face meets each circular face)
- Vertices: none

Example. Count the faces, edges and vertices of a sphere.

In fig-02 the sphere has only its curved surface. So:

- Faces: **1 curved face**
- Edges: none
- Vertices: none

Notice the difference: the cylinder has two curved edges and no vertex, while the sphere has no edge at all.

1. 1. How many circular faces does a solid cylinder have?
answer: Two.

2. 2. A cylinder has how many faces altogether?
answer: Three faces (two circular faces and one curved face).

3. 3. How many vertices does a sphere have?
answer: None — a sphere has no vertex and no edge.

4. 4. Name two properties that tell a cylinder apart from a sphere.
answer: A cylinder has two circular faces and two curved edges; a sphere has one curved surface and no edges.

Objective 2

Volume of Cubes, Cuboids and Prisms

EXPLANATION

The volume of a solid tells you how much space it takes up. You already know the units — when you measure in centimetres, volume is in cubic centimetres (cm^3), and when you measure in metres, it is in cubic metres (m^3).

For a cuboid, the volume is found by multiplying the area of its base by its height:

Volume of cuboid = area of base × height
= length × breadth × height
= lbh

Remember that length, breadth and height are all measured in the same unit, and you multiply them together.

A cube is a special cuboid where every edge is the same length, l. So all three dimensions equal l, and

Volume of cube = l × l × l = l^3

You can read l^3 as "l cubed".

EXPLANATION

The same idea — volume equals area of base × height — also works for a prism. A prism has two identical parallel faces joined by a constant cross-section. So for any prism:

Volume of prism = area of base × height

You first find the area of the base face, then multiply it by the height (the distance the prism runs).

WORKED_EXAMPLE

Example 1. Find the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.

Volume of cuboid = L × b × h

length = 3 cm, breadth = 2 cm, height = 8 cm

Volume = 3 cm × 2 cm × 8 cm
= 48 cm^3

So the cuboid has volume 48 cm^3.

WORKED_EXAMPLE

Example 2. A cuboid has volume 24 cm^3. Its base is 6 cm by 2 cm. Find its height.

Volume of cuboid = L × b × h

length = 6 cm, breadth = 2 cm, height = ?

Volume = 24 cm^3

24 cm^3 = 6 cm × 2 cm × h
24 cm^3 = 12 cm^2 × h

Divide both sides by 12 cm^2:

24 cm^3 ÷ 12 cm^2 = h
2 cm = h

So the height is 2 cm.

WORKED_EXAMPLE

Example 3. Find the volume of a cube of side 3 cm.

Volume of cube = l^3

$V = (3\text{ cm})^3$
$V = 3\text{ cm} \times 3\text{ cm} \times 3\text{ cm}$
$V = 27\text{ cm}^3$

So the cube has volume 27 cm^3.

WORKED_EXAMPLE

Example 4. Find the volume of the triangular prism. Its triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

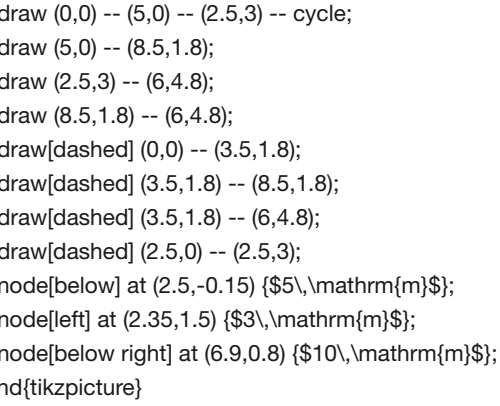
First find the area of the triangular base:

Area of triangle $= \frac{1}{2} \times b \times h$
$= \frac{1}{2} \times 5 \times 3$
$= 7.5\text{ m}^2$

Then volume = area of base × height
$= 7.5\text{ m}^2 \times 10\text{ m}$
$= 75\text{ m}^3$

So the prism has volume 75 m^3.

VISUAL



1. Calculate the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.
answer: 96 cm^3

2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
answer: 30 m^3

3. What is the volume of a cube of side 4 cm?
answer: 64 cm^3

4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m^3 of water. How deep is the water in the tank?
answer: 3 m

Result — per rubric dimension

Scores are 1–5. Every figure is the mean of 3 draws rated by 2 judge pass(es) — 6 judgments per cell.

Rubric	C0	C1	C2	C3	C1 – C0	C2 – C1	C3 – C2
Objective and curriculum alignment	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00
Taught-lesson alignment	4.00	4.00	4.33	4.00	+0.00	**+0.33**	**-0.33**
Scope and conceptual correctness	4.50	4.83	4.83	4.83	**+0.33**	+0.00	+0.00
Student-facing clarity and usefulness	4.83	5.00	5.00	5.00	**+0.17**	+0.00	+0.00

Why the judge scored it that way

Every rationale the judge wrote for this lesson, deduplicated across draws and passes. These are its words, not a summary.

Objective and curriculum alignment — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` Both lesson objectives are covered completely and in appropriate depth for JSS1 Third Term. The pack matches the curriculum target well, addressing properties of cylinders and spheres and volume of cube, cuboid, and prism.
- `C0` Both lesson objectives are fully addressed in the revision pack: properties of cylinders and spheres are described, and the volume formulae for cube, cuboid, and prism are explained and practised. The pack maps directly onto the JSS1 Third Term curriculum content.
- `C0` Both stated objectives are fully addressed in the revision content, worked examples, and practice questions. The pack maps directly onto the JSS1 Third Term curriculum targets for this lesson.
- `C0` Both lesson objectives are fully addressed with appropriate content, worked examples, and practice questions. The pack maps cleanly onto the stated curriculum targets for JSS1 Third Term.
- `C0` The revision pack comprehensively covers both stated objectives — identifying properties of cylinders and spheres, and finding volume of cube, cuboid and prism — with no significant gaps or additions that fall outside the lesson's curriculum target.
- `C0` All stated lesson objectives are fully addressed with appropriate depth for JSS1. Both the properties strand and the volume strand receive thorough treatment.
- `C1` The revision pack maps directly onto both stated lesson objectives and reproduces the teacher's evaluation questions, indicating strong alignment with the intended curriculum targets for this JSS1 Third Term lesson.
- `C1` The revision pack directly addresses both stated lesson objectives — properties of cylinders and spheres, and volume of cubes, cuboids and prisms — without missing any element and without straying beyond the lesson scope.
- `C1` Both stated lesson objectives are fully addressed in the correct order, and the content maps directly onto the JSS1 Third Term curriculum target for three-dimensional shapes. Nothing important is missing.
- `C1` The pack covers both stated lesson objectives completely and at the right JSS1 level, including the specific shapes (cube, cuboid, prism) and the properties (faces, edges, vertices) that the curriculum targets.
- `C1` Both stated lesson objectives are addressed completely and in order. The pack covers exactly what the curriculum target requires for JSS1 Third Term week 5 on 3D shapes II.
- `C1` Both lesson objectives are explicitly and completely addressed in the revision pack, with content and practice items that match the intended curriculum level and assessment tasks.
- `C2` The pack covers both stated lesson objectives completely and directly, matching the curriculum targets for JSS1 Third Term week 5 on 3D shapes and volume.
- `C2` The pack addresses both stated lesson objectives completely and directly, matching the curriculum target for JSS1 Third Term week 5 on three-dimensional shapes. No objective is missing or skimmed.
- `C2` Both lesson objectives are covered comprehensively and in the correct order. The pack addresses every shape and concept named in the objectives without omitting anything.
- `C2` The revision pack covers both stated lesson objectives completely and in the correct order. Nothing relevant to the objectives is omitted, and no extraneous objectives are introduced.

- `C2` The pack maps directly and completely onto both lesson objectives. Every concept the teacher intended students to learn is present and given appropriate treatment.
- `C2` Both stated lesson objectives are covered completely and in the correct JSS1 curriculum context. No objective is omitted or misrepresented.
- `C3` Both stated lesson objectives are covered completely and explicitly in the revision pack, with no objective left out or misrepresented. The pack maps cleanly onto the JSS1 curriculum target for this topic.
- `C3` Both stated lesson objectives are fully addressed in the correct depth for JSS1. Nothing essential from the objectives is missing, and the pack does not stray beyond the curriculum target.
- `C3` The revision pack addresses both stated objectives completely and in the correct order, with appropriate depth for JSS1. Nothing important from the objectives is omitted.
- `C3` The pack addresses every listed objective completely and in the correct curriculum context for JSS1 Third Term. Nothing from outside the lesson's scope is introduced at the objective level.
- `C3` Both stated lesson objectives are fully addressed in the correct order, and every assessment question from the teacher's plan is included. The pack leaves no objective uncovered.
- `C3` Both stated lesson objectives are fully addressed in the correct order, and the curriculum target for JSS1 Term 3 on 3D shapes and volume is well covered. There are no gaps or off-topic additions.

Taught-lesson alignment — C0 4.00, C1 4.00, C2 4.33, C3 4.00

- `C0` The pack reflects the lesson's sequence and emphasis faithfully. A minor gap is that the lesson stresses student identification through handling physical shapes, which a take-home pack cannot replicate, but the written content otherwise aligns well with what was taught and how it was assessed.
- `C0` The pack closely mirrors the lesson's sequence and emphasis, and the worked examples match what the teacher would have put on the board. A minor gap is that the pack does not replicate the interactive identification activity (students calling out properties as the teacher displays shapes), but this is appropriate for a take-home revision context.
- `C0` The pack faithfully follows the lesson's sequence and covers the same content and depth, though it cannot replicate the hands-on display and student-identification activities. The omission of the tank depth problem (reverse volume) is a minor gap, but overall alignment is strong.
- `C0` The pack faithfully reflects the lesson's sequence and emphasis, covering both strands in the right order and at the expected depth. The slight extension to a triangular prism is within the lesson's stated scope and does not misrepresent what was taught.
- `C0` The pack closely mirrors the lesson's sequence and depth, reusing the teacher's own examples and evaluation questions, though the interactive shape-identification activity is necessarily adapted into descriptive questions for home use.
- `C0` The pack closely mirrors the lesson's sequence and emphasis. The main minor gap is that the lesson plan's depth on prisms is minimal (one line), yet the pack gives prisms considerable attention — though this is still within the lesson objective. The duplication of the volume section is a structural inefficiency but does not misrepresent the lesson.
- `C1` The pack faithfully reflects the lesson's sequence and instructional emphasis. The only minor gap is that the pack does not replicate the hands-on display/identification activity with physical shapes, though this is hard to reproduce in a take-home document.
- `C1` The pack faithfully reflects the lesson's sequence and the teacher's own questions and methods; the only minor gap is that the sphere-property section goes slightly beyond what the lesson's set questions demanded, but this is a harmless extension rather than a misalignment.
- `C1` The pack faithfully mirrors the lesson's sequence, method, and depth. The only minor gap is that the pack does not replicate the hands-on display/identification activity where students interacted with physical shapes, but this is unavoidable in a take-home format and the content coverage is otherwise excellent.
- `C1` The pack closely reflects the lesson's sequence, emphasis, and depth; the only minor departure is the addition of a triangular prism worked example, which extends slightly beyond what the teacher explicitly demonstrated but is consistent with the prism objective.
- `C1` The pack follows the lesson sequence and depth closely and mirrors the teacher's evaluation questions. It slightly shifts the cylinder/sphere activity from a class oral exercise to a written self-check, which is a minor adaptation appropriate for home use but represents a small departure from the taught method.
- `C1` The pack closely mirrors the lesson's sequence, emphasis, and depth; the only minor gap is that sphere volume is not taught but that is intentional per the lesson plan, and the pack correctly omits it, so alignment is strong throughout.

- `C2` The pack closely reflects the lesson's sequence, method, and evaluation questions; it loses one point because the lesson presentation gave students a hands-on identification activity with physical shapes, which the pack cannot replicate, and the prism example goes slightly beyond the depth implied by the lesson notes.
- `C2` The pack closely reflects the lesson's sequence and teaching method, including the property-by-property examination of shapes and the formula-derivation approach to volume. The addition of a triangular prism example is a minor extension but is consistent with the stated objective on prisms.
- `C2` The pack mirrors the lesson's sequence, activities, and depth with high fidelity. The inclusion of the teacher's own in-class and evaluation questions confirms strong alignment with what was actually taught and assessed.
- `C2` The pack faithfully reflects the lesson's sequence, method and emphasis. It loses one point because the teacher's presentation was activity-based (students identifying properties from displayed shapes), but the pack is necessarily text-based and cannot fully replicate that hands-on element — a minor but real difference in mode.
- `C2` The pack closely mirrors the lesson's sequence, method, and depth. It loses one mark because the lesson presentation emphasised student identification of properties through handling physical shapes and board discussion, while the pack is entirely expository — the interactive, discovery-based element of the classroom is naturally absent but its framing could have been made more activity-like for home use.
- `C2` The pack's structure, content, and practice questions align tightly with how the lesson was planned and sequenced. The emphasis and depth match what the teacher intended.
- `C3` The pack mirrors the lesson's sequence, method (formula derivation then worked examples), and evaluation questions closely. The minor shortfall is that the classroom activity element — students physically identifying shapes and contributing properties — cannot be replicated in a written pack, but the pack compensates well with explanatory prose.
- `C3` The pack faithfully reflects the lesson's sequence, method, and emphasis. It loses one point because the lesson's interactive element (students identifying properties from physical shapes, class discussion) is naturally absent in a take-home format, but the pack compensates well with a descriptive introduction and worked example.
- `C3` The pack faithfully reproduces the lesson's sequence and emphasis. A minor gap is that the pack does not explicitly reference the use of physical shapes/manipulatives that the teacher used, but this is expected in a text-based home revision pack.
- `C3` The sequence, emphasis, and method align closely with the taught lesson; the additional worked examples enrich rather than distort the lesson's intent, keeping the pack well-aligned overall.
- `C3` The revision pack closely tracks the lesson's sequence, emphasis, and method. A minor gap is that the lesson's interactive element — students identifying properties from physical shapes — cannot be reproduced in a written pack, but everything else aligns well.
- `C3` The pack closely tracks the lesson's sequence, key activities, and the teacher's own evaluation questions. It loses one point because it adds a triangular prism worked example with numbers not used in the lesson, going slightly beyond the depth demonstrated in the teacher's presentation.

Scope and conceptual correctness — C0 4.50, C1 4.83, C2 4.83, C3 4.83

- `C0` The content is mathematically correct throughout, and the scope mostly matches the lesson objectives. The triangular prism example is within the stated objective but goes a step further in detail than the teacher's own presentation, which is a very minor concern rather than an error.
- `C0` There are no mathematical errors in the pack; all formulae, worked examples, and answers in the answer key are correct. The content stays within the lesson boundary, though the brief note linking cuboids to the general prism concept is a small conceptual extension that JSS1 students may not need.
- `C0` There are no conceptual or computational errors throughout the pack. The content stays within JSS1 scope and correctly handles all the mathematics required by the lesson objectives.
- `C0` The mathematical content is accurate and stays within the lesson boundary. The treatment of 'curved edges' on a cylinder is a minor pedagogical nuance rather than a conceptual error, and does not mislead students at this level.
- `C0` The pack stays firmly within the lesson's scope and contains no mathematical or conceptual errors; every formula, property description, and numerical answer has been verified as correct.
- `C0` There are no mathematical or conceptual errors in the pack. The content stays within the expected JSS1 scope and all answers in the check-your-work section are verified correct.
- `C1` There are no mathematical or conceptual errors, and the pack does not stray outside the scope of the lesson. The treatment is appropriate for JSS1 depth.
- `C1` There are no mathematical or conceptual errors anywhere in the pack, and it remains correctly scoped to what a JSS1 lesson on this topic is expected to cover.

- `C1` The pack stays strictly within the lesson boundary (no extraneous shapes or advanced content introduced) and contains no conceptual or arithmetic errors. The note about 'faces' vs. 'curved surface' in the answer key is handled precisely.
- `C1` There are no mathematical or conceptual errors in the pack; every formula, worked example, and answer key entry is accurate and appropriately scoped for JSS1.
- `C1` There are no conceptual or mathematical errors. The triangular prism worked example is a minor scope extension beyond what the teacher's plan explicitly demonstrated, but it is correct and directly supports the stated objective.
- `C1` There are no mathematical errors or conceptual inaccuracies, and the content stays firmly within the expected lesson boundary without going beyond what JSS1 students are expected to know at this stage.
- `C2` There are no conceptual or arithmetic errors anywhere in the pack, and the content stays within the expected boundary of JSS1 three-dimensional shapes and volume work.
- `C2` There are no mathematical or conceptual errors in the pack. The scope stays within JSS1 expectations and does not introduce content beyond what is required by the lesson objectives.
- `C2` There are no mathematical errors and no conceptual mistakes. The pack stays strictly within the lesson's scope — it introduces nothing beyond what the lesson planned, and every formula and answer is correct.
- `C2` Conceptually and mathematically the pack is accurate throughout. The only mild concern is the addition of the triangular prism example, which stretches the scope a little beyond what the lesson plan explicitly required, though it is a legitimate and helpful illustration of the prism formula.
- `C2` There are no conceptual or computational errors anywhere in the pack, and the content stays within the expected JSS1 lesson boundary without straying into out-of-scope material.
- `C2` There are no mathematical or conceptual errors in the pack. All content stays within the intended lesson boundary and is correct at JSS1 level.
- `C3` There are no conceptual or computational errors anywhere in the pack, and it stays strictly within the boundaries of what a JSS1 lesson on this topic should cover — no unnecessary extension or out-of-scope content.
- `C3` There are no conceptual or computational errors anywhere in the pack, and the content stays strictly within the lesson boundary — no extraneous solids or formulas are introduced.
- `C3` There are no mathematical or conceptual errors, and the pack does not stray outside the lesson boundary. All definitions, formulas, and calculations are correct and grade-appropriate.
- `C3` Every mathematical statement, formula, and worked answer in the pack is conceptually and numerically correct, and the content stays within the JSS1 lesson boundary without straying into out-of-scope territory.
- `C3` The content is conceptually and mathematically accurate throughout. The triangular prism example is a small extension beyond the lesson's explicit worked content but remains within the stated objective boundary and is handled correctly.
- `C3` Every mathematical statement and calculation in the pack is correct, and the content stays within the expected JSS1 scope for this topic — no errors or out-of-level concepts are introduced.

Student-facing clarity and usefulness — C0 4.83, C1 5.00, C2 5.00, C3 5.00

- `C0` The pack is exceptionally well-suited for independent home use by JSS1 students: language is simple and age-appropriate, examples are relatable, steps are explicit, and full worked answers are provided so students can self-check without a teacher present.
- `C0` The pack is well-structured and written at an appropriate level for JSS1 learners, with familiar everyday objects used as anchors, step-by-step examples, and complete solutions — making it genuinely useful as independent take-home revision material.
- `C0` The pack is very well structured for independent home use: accessible language, concrete analogies, scaffolded worked examples, and fully explained answers make it highly usable for a JSS1 student revising alone.
- `C0` The pack is exceptionally well-structured for independent home use: relatable contexts, step-by-step examples, and detailed answer explanations allow a JSS1 student to work through it confidently without teacher support.
- `C0` The pack is generally very accessible and well-structured for JSS1 home use, with good use of familiar examples and clear worked solutions; the main weakness is the unexplained duplication of content, which could confuse an unsupported learner.
- `C0` The pack is exceptionally well-suited for independent home revision: the language is age-appropriate, the worked examples are clear and well-stepped, and the self-check answers allow learners to verify and understand their work without a teacher.
- `C1` The pack is well-structured, uses familiar language and relatable contexts, and provides sufficient scaffolding for a JSS1 student to revise independently and verify their own work.

- **C1** The pack is clearly written, well-structured, and includes sufficient explanation and answer keys for a JSS1 student to review the lesson meaningfully on their own at home.
- **C1** The language is simple, concrete, and well-structured for a 12–13-year-old working alone at home. The inclusion of explained answers makes it genuinely self-correcting and pedagogically useful as independent revision material.
- **C1** The language is simple and appropriately pitched for JSS1; the structure—explanation, worked examples, practice, answers—makes the pack highly usable as standalone take-home revision material.
- **C1** The pack is well-structured, uses relatable everyday examples, and provides both worked examples and answer keys — everything a JSS1 student needs to revise independently at home without teacher support.
- **C1** The pack is exceptionally well-structured for independent home use: real-world contexts, clear language, fully worked examples, and explained answer keys combine to make it highly usable for a JSS1 learner revising alone.
- **C2** The pack is exceptionally well-structured for independent home use: clear language, fully worked models, and self-check answers with reasoning make it highly useful for a JSS1 learner revising alone.
- **C2** The pack is well-suited for independent home use: language is simple and relatable, the progression from explanation to example to practice is logical, and the answer key gives enough working for a student to self-diagnose errors without a teacher present.
- **C2** The pack is well-structured, uses simple language appropriate for JSS1, and provides worked examples alongside practice questions with answers, making it highly usable as a standalone take-home revision tool.
- **C2** The pack is exceptionally well-suited for independent home use: language is clear and age-appropriate, steps are explicit, figures are described, and self-checking is built in. A JSS1 learner could work through it without any outside help.
- **C2** The pack is exceptionally well suited as independent take-home material: the language is accessible, the structure is logical, and the self-check answers with explanations allow unsupported learners to identify and correct their own mistakes.
- **C2** The language is accessible and age-appropriate, the layout moves from explanation to example to practice to answers in a logical self-study flow. A JSS1 student working alone at home would find this pack genuinely usable.
- **C3** The pack is exceptionally well-structured for independent home use: plain language, relatable contexts, step-by-step solutions, and a full answer key with reasoning make it highly accessible and useful for JSS1 learners working alone.
- **C3** The pack is exceptionally well structured for independent home use: accessible language, logical progression, multiple worked examples, matching practice questions, and fully explained answers make it highly useful for a JSS1 student revising without a teacher present.
- **C3** The pack is exceptionally well-structured for independent home use: the layout progresses logically from explanation to example to practice to self-check, and the language is accessible to a JSS1 learner working without a teacher.
- **C3** The pack is exceptionally well-structured for independent home use: everyday contexts, stepwise examples, and fully explained answers give a JSS1 student everything needed to review, practise, and self-assess without a teacher present.
- **C3** The pack is well-structured, uses simple language appropriate for JSS1, and provides enough scaffolding — analogies, worked examples, practice questions, and annotated answers — for a student to review the lesson effectively without a teacher present.
- **C3** The pack is written in simple, direct language appropriate for JSS1, progresses logically from explanation to example to practice, and provides complete answer keys — making it highly suitable as a standalone take-home revision resource.